EXPERIMENT 28

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CODE:

```
clc;
clear;
close all;
% Data Bits
data = [1 0 1 1 0];
data = 2*data - 1;
% Convert to ±1
% PN Sequence (Length = 5)
pn = [1 -1 1 -1 1];
% DSSS Spreading
spread = kron(data,pn);
% Despreading
rec = reshape(spread,length(pn),[]);
rec_data = sign(sum(rec));
rec_data = (rec_data+1)/2;   % Convert back to binary
% Display
disp('Original Data:');
disp((data+1)/2);
disp('Recovered Data:');
disp(rec_data);
% Plots
figure;
```

```
subplot(3,1,1)
stem((data+1)/2,'filled')
title('Original Data')
grid on
subplot(3,1,2)
stem(spread,'filled')
title('DSSS Spread Signal')
grid on
subplot(3,1,3)
stem(rec_data,'filled')
title('Recovered Data')
grid on
```

OUTPUT:

